

Model Predictive Variable Impedance Control of Manipulators for Adaptive Precision-Compliance Tradeoff

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I. INTRODUCTION

Abstract—Impedance control (IC) is widely used in contact-rich manipulator tasks since it provides manipulators with both operation precision and contact compliance, and trades them off by impedance parameters. However, fixed impedance parameters limit the applicability of IC in complex tasks during which the focus on the operation precision or the contact compliance is variable, such as physical human–robot collaboration and complex assembly tasks. This article presents model predictive variable impedance control approaches for adaptive precision-compliance tradeoff to satisfy variable task requirements. Specifically, we establish a novel impedance model, which transforms the variable impedance law design problem into a control law design problem, and allows us to consider novel impedance constraints that can determine manipulators' extreme precision and compliance properties. According to whether state constraints are further considered, one-step (OS) and multistep (MS) model predictive control approaches are proposed to solve these transformed control problems, and the corresponding optimization problem of the OS MPC can be established as a QP problem due to the special form of the novel impedance model. A tank-based approach is further proposed to correct the variable impedance laws for the system passivity concern. Various comparative experiments are conducted to validate the effectiveness of the presented approaches.

Index Terms—Model predictive control, optimization, precision-compliance tradeoff, variable impedance control (VIC).

IMPEDANCE control (IC) [1] is a reliable control architecture for contact-rich robot tasks since it provides robots with the contact-compliance property. Actually, IC determines a tradeoff between the operation precision and contact compliance for robots. The specific tradeoff mode is dependent on the impedance parameters of the IC. However, a single and fixed tradeoff mode is only effective for structure tasks, such as those in factory assembly lines. For more flexible tasks, such as working with humans or other robots, robots are expected to have different precision-compliance tradeoff modes to cope with different task statuses [2]. Variable impedance control (VIC) which combines the IC with a variable impedance law serves as a good solution to provide robots with this adaptive ability [3]. Thus, in recent years, numerous works [4], [5], [6] have been done to explore the benefits of the VIC.

The adaptive ability is fully determined by the variable impedance law which is the key element of the VIC. Design of the variable impedance law is not trivial and goes beyond the control engineering domain [7], [8]. A popular way to establish the variable impedance law is based on human demonstration. In [9], an interface that allows a human teacher to indicate impedance variation by physically interacting with the robot during the task execution was presented. In [10], the interface presented in [9] was further enhanced, and a modified Gaussian mixture model/Gaussian mixture regression (GMM/GMR) learning algorithm was designed to transmit the human impedance variation law to the robot. In [11], the variable impedance law was described as a dynamic movement primitive model in which the parameters were optimized by applying the locally weighted regression algorithm on a demonstrated dataset. With the same idea of using supervised learning algorithms as [10] and [11], the variable impedance law was derived in [12] using the hidden semi-Markov model/Gaussian mixture regression algorithm. A major disadvantage of the above supervised-based demonstration learning algorithms is that the ground truth of the human impedance parameters is difficult to be obtained. For example, the electromyographic (EMG) signal used to represent human stiffness in [11] requires a complex preprocessing on the raw EMG signal. An approach to deal with this issue was presented in [13], where human impedance parameters were served as hidden variables of a Gaussian process model.

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Demonstration learning-based VIC aims to transform the human impedance variable skills to robots, where there may not exist specific task-related objective functions. For some tasks, we may have explicit objectives, such as increasing tracking precision and improving contact compliance for contact-rich trajectory tracking tasks. With an objective function, we can compare two variable impedance laws and define the optimal variable impedance law. Recently, reinforcement learning (RL) algorithms have become popular for designing optimal variable impedance laws for these tasks with explicit objectives. By designing reward functions to be consistent with the task objectives, RL algorithms always derive helpful variable impedance laws [14], [15], [16], [17]. However, since RL algorithms are always trained in virtual environments and improve their performances through trial-and-errors, the final performances of learned variable impedance laws are determined by gaps between the virtual and real environments. The reason for trial-and-errors is that RL algorithms always assume that the environment model is completely or partly unknown, which is the major difference between RL algorithms and control algorithms. Model predictive control (MPC) is the most similar to RL algorithms among all control algorithms, either from the perspective of formulation or objective [18], [19].

MPC-based VIC approaches have also been reported in recent years. Similar to the RL-based VIC approaches, the MPC-based VIC approaches also aim to search the optimal variable impedance laws with respect to task-specific objective functions. Moreover, MPC-based VIC approaches are more efficient and interpretable than RL-based VIC approaches, due to their full utilization of the model knowledge. In [20], [21], and [22] where trajectory tracking tasks were considered, impedance models were rewritten as linear models, and MPC approaches with quadratic cost functions were presented to online generate optimal impedance parameters for robots to trade off the tracking precision against the contact compliance. In [23] and [24], LQR approaches, i.e., unconstrained MPC approaches with the quadratic cost function and infinite predictive horizon, were used to derive variable impedance laws for compliant physical human-robot interactions. However, all of the aforementioned MPC-based VIC approaches did not consider constraints since they need to maintain solutions of the MPC optimization problems to be consistent with the formulation of the impedance controller. Furthermore, with linear models used in previous MPC-based VIC approaches, the precision-compliance tradeoff problem was only partially solved, i.e., cost functions of the MPC approaches were not exactly consistent with the precision-compliance tradeoff objective.

In this article, novel MPC-based VIC approaches, which can consider constraints and exactly handle the adaptive precision-compliance tradeoff problem are presented. Specific contributions are listed as follows:

- 1) A novel impedance model, in which impedance parameters are directly served as the control input, is established to replace the classical linear impedance model. With this novel impedance model, the variable impedance law design problem can be naturally transformed into a controller design problem, and we can design the MPC

cost function to exactly capture the objective of robot precision-compliance tradeoff.

- 2) Due to the specificity of the novel impedance model, we can accept any form of the optimal solution of the MPC optimization problem, even with the numerical form. Constraints are thus acceptable in the MPC optimization problem. Novel constraints, named impedance constraints, which rule the scope of the impedance parameters and determine robots' extreme precision and compliance properties, are considered to improve robot operation performances.
- 3) Two specific MPC approaches, i.e., one-step (OS) MPC and multistep (MS) MPC approaches are presented to design the variable impedance laws for different types of tasks. The OS MPC optimization problem satisfies the standard QP formulation, and it is designed to solve tasks with only impedance constraints. The MS MPC approach faces the heavier computational burden but is free of the limitation for constraint formulation.
- 4) A tank-based approach is presented to ensure the passivity of the variable impedance systems resulting from the presented MPC-based VIC approaches, which can increase the robot operation safety for contact-rich tasks.

Various practical experiments are conducted to validate the effectiveness of the presented MPC-based VIC approaches. The remaining contents are structured as follows. The background of the robot dynamic control and impedance control are introduced in Section II. A detailed description of the studied problem is given in Section III. MPC-based VIC approaches and passivity analyses are introduced in Sections IV and V, respectively. Experimental results are shown in Section VI. A discussion is provided in Section VII. Finally, Section VIII concludes this article and presents the future works.

II. CONTROL BACKGROUND

A. Robot Dynamic Control

We consider the following n -DOF robot dynamic model:

$$H(q)\ddot{q} + C(q, \dot{q})\dot{q} + g(q) = J^T f + \tau \quad (1)$$

where $H(q) \in R^{n \times n}$, $C(q, \dot{q}) \in R^{n \times n}$, and $g(q) \in R^n$ are the inertia matrix, Coriolis/centripetal matrix, and gravity vector of the robot, respectively. Matrix $J(q) \in R^{m \times n}$ is the robot Jacobian and the vector $f \in R^m$ is the contact force where $m < n$ is the dimension of the task space. To track the desired joint acceleration ν , the controller can be designed as

$$\tau = H(q)\nu + C(q, \dot{q})\dot{q} + g(q) - J^T f. \quad (2)$$

The velocity in task space and joint space can be related as $\dot{\xi} = J\dot{q}$ where $\xi = [x \ y \ z \ \theta_x \ \theta_y \ \theta_z]^T$ is the pose of the end-effector in task space. Taking the derivative of both sides, one can obtain

$$\nu = J^+ \left(\ddot{\xi} - \dot{J}\dot{q} \right). \quad (3)$$

Substituting (3) into (2), one can obtain the following control law to track the desired task space acceleration v

$$\tau = H(q)J^+ \left(v - \ddot{q} \right) + C(q, \dot{q})\dot{q} + g(q) - J^T f \quad (4)$$

B. Impedance Control

The impedance control in task space can be described as

$$M\ddot{e} + D\dot{e} + Ke = f \quad (5)$$

where M , D , and K are desired inertia, damping, and stiffness matrices, respectively. Assuming M , D , and K are positive definite and diagonal, we can decouple the model (5) to obtain the following 1-D model:

$$m_j\ddot{e}_j + d_j\dot{e}_j + k_je_j = f_j \quad (6)$$

where the subscript j indicates the j th dimension, $e_j = \xi_j - \xi_{r,j}$ and $\xi_{r,j}$ is the j th dimension of the reference pose. For convenience, we only consider the 1-D model, and the subscript j is omitted in the rest of the article. The impedance model (6) is actually a mass-damping-spring system, and describes the relationship between the contact force f and the robot's motion behavior ξ , such as

$$\ddot{\xi} = \frac{f - ke - d\dot{e}}{m} + \ddot{\xi}_r. \quad (7)$$

We can set the v in (4) as $v = \ddot{\xi}$ and compute the final control signal τ as follows:

$$\begin{aligned} \tau = H(q)J^+ \left(\frac{f - ke - d\dot{e}}{m} + \ddot{\xi}_r - \ddot{q} \right) \\ + C(q, \dot{q})\dot{q} + g(q) - J^T f. \end{aligned} \quad (8)$$

The relationship (7) implies that the goal of the impedance control is not to eliminate the tracking error $(e, \dot{e})$, but to find a tradeoff between the tracking precision and the contact compliance, i.e., the influence to the system caused by the contact force f . The contact compliance is important to ensure the contact safety for contact-rich tasks, and thus, we cannot ignore it. The specific precision-compliance tradeoff mode is determined by the impedance parameters m , d , and k . For example, large stiffness and damping parameters (k, d) will reduce the system compliance but can eliminate the tracking error $(e, \dot{e})$ well.

III. PROBLEM STATEMENT

We have mentioned that the impedance control involves a tradeoff between the tracking precision and contact compliance. However, a pair of fixed impedance parameters is related to one particular tradeoff mode. For different tasks, or even for different stages of a task, we may need different tradeoff modes. For example, the robot is expected to have large stiffness when it is asked to track a high-speed trajectory, and have low stiffness when the reference trajectory's speed is low. Although, a robot with large stiffness can also track the low-speed reference trajectory well, it could be unsafe when the robot's working space is

unstructured, such as working near humans. Thus, an approach to online adjust tradeoff modes, i.e., determine different pairs of impedance parameters according to the task status, is very significant. These approaches aiming to find proper variable impedance laws and taking them into the impedance control framework are referred to as the VIC.

In this work, we consider MPC-based approaches to online adjust the precision-compliance tradeoff mode. We refer to these approaches as MPC-based VIC approaches. MPC is a popular and convincing approach due to its strong interpretability and optimization background. Previous MPC-based VIC approaches [20], [21], [22], [23], [24] approximated the model (6) as the following linear state space model (or similar to this linear state space model):

$$\begin{cases} \dot{s} = As + bw \\ s = [e \quad \dot{e} \quad f]^T \\ A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \\ b = [0 \quad 1 \quad 0]^T \\ \omega = \left[\frac{-k}{m} \quad \frac{-d}{m} \quad \frac{1}{m} \right]^T s. \end{cases} \quad (9)$$

The gap between the models (9) and (6) is caused by the assumption $\dot{f} = 0$. Discretizing the model (9) and incorporating it in the MPC framework result in the following MPC optimization problem:

$$\min_{\Upsilon_t} J(\Upsilon_t) = \left(\sum_{i=t}^{t+H-1} \|s_i\|_{Q_t}^2 + r_t\omega_i^2 \right) + \|s_{t+H}\|_{Q_t}^2 \quad (10)$$

subject to

$$s_{i+1} = (AT + I)s_i + Tbw_i \quad (11)$$

where H is the prediction horizon, Υ_t represents the control sequence $(\omega_t, \dots, \omega_{t+H-1})$, (11) is the discretized model, and T is the control period. Q_t and r_t are weight parameters of the cost function. Since the model (11) is linear and the cost function (10) is quadratic, the optimal solution ω_t^* satisfies the form

$$\omega_t^* = \beta_t^* s \quad (12)$$

where β_t^* can be computed by Ricatti recursion. By consisting the last equation in (9) with the solution form (12), we can compute the optimal impedance parameters. Solving the optimization problem (10), (11) at every time step, we can obtain an optimal variable impedance law.

Observing the cost function (10), we can find that the previous MPC-based VIC approaches actually adjust the tradeoff between the tracking error $(e, \dot{e})$ and the control input ω_t . Although the control input ω_t somehow reflects the degree of the compliance due to its special form (12), they are still not the same. This drawback is caused by the formulation of the model (11), in which impedance parameters do not directly appear. Additionally, if we further consider other constraints, such as constraints on state s , the specific form (12) will become unobtainable.

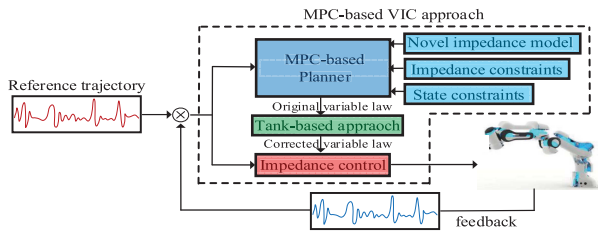


Fig. 1. Scheme of the proposed model predictive variable impedance control approach.

Thus, the first problem that will be considered in this article is to derive a novel impedance model, which allows us to handle the precision-compliance tradeoff problem more directly and conveniently.

Secondly, we consider injecting the human domain knowledge into the tradeoff adjustment process to make it more effective. In this article, we will introduce impedance constraints that define the scope of the impedance parameters into the MPC optimization problem. Impedance constraints have never been considered by the previous MPC-based VIC approaches since impedance parameters are hidden in the control input ω in the model (9). However, impedance constraints are significant to ensure some extreme properties. For example, if the minimal stiffness and damping parameters are set as large values, we can ensure the robot's high tracking precision. We also consider constraints on the state s . Though these state constraints can also be considered in previous MPC-based VIC approaches in which the optimization problems like (10) and (11), they never do it otherwise the ideal form (12) might be lost.

The final consideration is about the passivity issue. It has been shown that introducing a variable impedance law into the system (6) will raise potential nonpassive risks [25], [26]. Passivity is directly related to the safety and thus cannot be ignored. We consider using a tank-based approach, which will correct the variable impedance law derived by the MPC approach to ensure the passivity of the final system.

As a conclusion, the considered problems which have not been solved by previous MPC-based VIC approaches contain the following:

- 1) establishing a novel impedance model which allows us to consider the precision-compliance tradeoff problem more directly and conveniently;
- 2) introducing impedance and state constraints into the MPC optimization problem to make the variable impedance law more effective;
- 3) designing MPC-based VIC approaches with impedance and state constraints based on the novel impedance model;
- 4) using the tank-based approach to ensure the passivity of the variable impedance systems resulting from the MPC-based VIC approaches.

The overall scheme of the proposed MPC-based VIC is shown in Fig. 1. An original variable impedance law is firstly generated by the MPC-based planner with consideration of the impedance and state constraints according to the novel impedance model. The original variable impedance law is then corrected using

the tank-based approach to ensure the system passivity. The corrected variable impedance law is finally fed to the impedance controller to generate suitable control input for the manipulator.

IV. MPC-BASED VIC DESIGN

In this section, we firstly establish the novel impedance model, which allows us to conveniently handle the precision-compliance tradeoff problem. Then, we introduce the MPC-based VIC approaches based on this novel impedance model with considerations of impedance and state constraints.

As previously mentioned, the main drawback of the original model (9) or (11) is that the impedance parameters are hidden in the control input ω , and thus they cannot be directly used to construct the cost function conveying the idea of a proper precision-compliance tradeoff. Another drawback is that with this model, we cannot casually introduce constraints into the MPC optimization problem, otherwise the ideal form (12) may be destroyed. Taking these two issues in the mind, we slightly modify the original model (11) to make the impedance parameters directly serve as the control input. The novel impedance model is given as follows:

$$s_{t+1} = (AT + I)s_t + T(b + \beta)s_t^T u_t \quad (13)$$

where $\beta = [\frac{T}{2} \ 0 \ 0]^T$, and the input u_t is defined as

$$u_t = \left[-\frac{k_t}{m_t} \quad -\frac{d_t}{m_t} \quad \frac{1}{m_t} \right]^T. \quad (14)$$

Remark 1: We here introduce an additional vector β to allow the input u_t to influence the next time error e_{t+1} . Actually, with this setting, the model (9) is discretized with the uniform acceleration assumption.

With the above novel impedance model, the variable impedance law design problem is naturally transformed into a controller design problem. Moreover, the abovementioned two drawbacks can be overcome. Firstly, the input u which is constructed by impedance parameters and directly related to the contact compliance can be used to construct the cost function. Secondly, we can introduce other constraints into the MPC optimization problem since we can accept any form of the optimal solution u_t^* , even the numerical form solution.

However, to get these excellent properties, the linear property is lost in the novel impedance model (13), which increases the difficulty for solving the corresponding MPC optimization problems. In the rest of this section, we will introduce two kinds of MPC-based VIC approaches, i.e., one-step (OS) MPC-based VIC approach and multistep (MS) MPC-based VIC approach. The reason to distinguish the OS and MS MPC-based VIC approaches is that these two approaches have their own strengths. For the OS MPC-based VIC approach, the optimization problem can be transformed into a QP problem in which the global optimal solution can be efficiently searched. However, this approach can only handle tasks with impedance constraints. On the contrary, the MS MPC-based VIC approach is accessible to tasks with broader types of constraints but with the expense of losing the exquisite form of the corresponding optimization problem.

A. One-Step MPC-Based VIC Design

For the one-step (OS) MPC-based VIC approach, we consider the following optimization problem at every time step:

$$\min_{u_t} J(u_t) = s_{t+1}^T Q_t s_{t+1} + u_t^T R_t u_t \quad (15)$$

subject to

$$\begin{cases} s_{t+1} = (AT + I)s_t + T(b + \beta)s_t^T u_t \\ u_t \in \Omega_t \end{cases} \quad (16)$$

where Q_t and R_t are positive-definite weight matrices, and Ω_t is the constrained set for the current input u_t . The cost function (15) implies a tradeoff between the tracking precision (related to minimize $s_{t+1}^T Q_t s_{t+1}$) and contact compliance (related to minimize $u_t^T R_t u_t$) because decreasing $s_{t+1}^T Q_t s_{t+1}$ usually needs a relatively large $u_t^T R_t u_t$. The input constraint $u_t \in \Omega_t$ is referred to as impedance constraints and it is formulated as

$$\begin{cases} \bar{G}_t u_t \preceq \bar{h}_t \\ \bar{A}_t u_t = \bar{b}_t. \end{cases} \quad (17)$$

The impedance constraints form (17) is very flexible. In this article, we consider two practically useful types of impedance constraints which both satisfy the formulation (17). The first type is called over-damping impedance constraints, which is listed as follows:

$$\begin{cases} m_t = c_t \\ k_t \in [k_{t,\min}, k_{t,\max}] \\ d_t = 2\sqrt{m_t} \frac{k_t}{\sqrt{k_{t,\min}}} \end{cases} \quad (18)$$

Constraints (18) result in a constant inertia, over-damping, and variable stiffness system. The over-damping property can improve the stability of the impedance system and is caused by the fact

$$d_t = 2\sqrt{m_t} \frac{k_t}{\sqrt{k_{t,\min}}} \geq 2\sqrt{k_t m_t}. \quad (19)$$

In this setting, the values of $\bar{G}_t$, $\bar{h}_t$, $\bar{A}_t$, and $\bar{b}_t$ are given as

$$\begin{cases} \bar{G}_t = \begin{bmatrix} -c_t & 0 & 0 \\ c_t & 0 & 0 \end{bmatrix} \\ \bar{h}_t = [k_{t,\max} \quad -k_{t,\min}]^T \\ \bar{A}_t = \begin{bmatrix} 0 & 0 & c_t \\ \frac{2\sqrt{c_t}}{\sqrt{k_{t,\min}}} & -1 & 0 \end{bmatrix} \\ \bar{b}_t = [1 \quad 0]^T \end{cases} \quad (20)$$

The second type of impedance constraints is called free-damping impedance constraints which have the form

$$\begin{cases} m_t = c_t \\ k_t \in [k_{t,\min}, k_{t,\max}] \\ d_t \in [d_{t,\min}, d_{t,\max}] \end{cases} \quad (21)$$

Compared with the over-damping impedance constraints, the free-damping impedance constraints decouple the damping parameter with the stiffness and inertia parameters. This setting will be useful when the damping variation is used to achieve special goals, such as reducing the robot velocity and maintaining the robot position. In this setting, the values of $\bar{G}_t$, $\bar{h}_t$, $\bar{A}_t$,

and $\bar{b}_t$ are given as

$$\begin{cases} \bar{G}_t = \begin{bmatrix} -c_t & 0 & 0 \\ c_t & 0 & 0 \\ 0 & -c_t & 0 \\ 0 & c_t & 0 \end{bmatrix} \\ \bar{h}_t = [k_{t,\max} \quad -k_{t,\min} \quad d_{t,\max} \quad -d_{t,\min}]^T \\ \bar{A}_t = [0 \quad 0 \quad c_t]^T \\ \bar{b}_t = 1. \end{cases} \quad (22)$$

Substituting the novel impedance model (13) into the objective function (15) and using the impedance constraints form (17), the OS MPC optimization problem can be rewritten as

$$\min_{u_t} J(u_t) = \frac{1}{2} u_t^T \bar{Q}_t u_t + \bar{p}_t^T u_t + \bar{c}_t \quad (23)$$

subject to

$$\begin{cases} \bar{G}_t u_t \preceq \bar{h}_t \\ \bar{A}_t u_t = \bar{b} \end{cases} \quad (24)$$

where

$$\begin{cases} \bar{Q}_t = 2T^2 s_t (b + \beta)^T Q_t (b + \beta) s_t^T + 2R_t \\ \bar{p}_t = 2T s_t (b + \beta)^T Q_t (AT + I) s_t \\ \bar{c}_t = s_t^T (AT + I)^T Q_t (AT + I) s_t. \end{cases} \quad (25)$$

Obviously, $\bar{Q}_t$ is positive-definite and, thus, the optimization problem (23) and (24) is strictly convex, and actually a QP problem which we can solve efficiently. It should be noted that the form of u_t^* is not limited since whatever the u_t^* form is, we can always recover the optimal impedance parameters (m_t^* , d_t^* , k_t^*) according to the u_t definition (14).

It should also be noted that weight matrices Q_t and R_t , or impedance constraint parameters (c_t , $k_{t,\min}$, $k_{t,\max}$) can all be set as s_t -dependent without changing the QP form. Using this property, we can insert domain experiences to the MPC-based VIC to improve algorithm's effectiveness. For example, when we observe large external force at time step t , we can decrease $k_{t,\max}$ or increase R_t to improve contact-compliance. The shortcoming for the OS MPC method is that it cannot consider state constraints, such as position and velocity constraints, since introducing these constraints may raise an infeasible QP problem. We leave state constraints considerations to the MS MPC method.

B. Multistep MPC-Based VIC Design

For the MS MPC-based VIC approach, we consider the following optimization problem at time step t :

$$\min_{U_t} J(U_t) = \sum_{i=t}^{t+H-1} l(s_i, u_i) + l_f(s_{t+H}) \quad (26)$$

subject to

$$\begin{cases} s_{i+1} = (AT + I)s_i + T(b + \beta)s_i^T u_i \\ s_i \in S_i \\ u_i \in \Omega_i \end{cases} \quad (27)$$

for all $i \in [t, t + H - 1]$ where $U_t = [u_t, \dots, u_{t+H-1}]$ is the control sequence; $l(s_i, u_i) = s_i^T Q_i s_i + u_i^T R_i u_i$ and $l_f(s_{t+H}) = s_{t+H}^T Q_{t+H} s_{t+H}$ are stage and terminal cost functions,

respectively. Scalar $H \geq 2$ is the prediction horizon; $s_i \in S_i$ and $u_i \in \Omega_i$ are state and impedance constraints, respectively. The impedance constraints here are set as the free-damping impedance constraints as

$$\begin{cases} m_i = c_i \\ k_i \in [k_{i,\min}, k_{i,\max}] \\ d_i \in [d_{i,\min}, d_{i,\max}]. \end{cases} \quad (28)$$

Noticed that the d_i is not coupled with k_i and m_i here because d_i is important to ensure the state constraints so that we need to give it enough variation freedom. For example, when the reference velocity is zero, increasing d_i will decrease robot velocity and prevent robot motion. Obviously, the optimization problem (26) and (27) is not a QP problem, and even not a convex optimization problem; we here simplify it using the variable-change and barrier function techniques.

We firstly introduce the variable-change technique. Since k_i and d_i should satisfy the constraints $k_i \in [k_{i,\min}, k_{i,\max}]$ and $d_i \in [d_{i,\min}, d_{i,\max}]$, respectively, we reparameterize k_i and d_i using $\mu_i = [\mu_{i,k} \ \mu_{i,d}]^T$ as

$$\begin{cases} k(\mu_{i,k}) = \frac{\pi(\mu_{i,k})+1}{2}(k_{i,\max} - k_{i,\min}) + k_{i,\min} \\ d(\mu_{i,d}) = \frac{\pi(\mu_{i,d})+1}{2}(d_{i,\max} - d_{i,\min}) + d_{i,\min} \end{cases} \quad (29)$$

where $\pi(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} \in (-1, 1)$. Thus, we have no constraints on the new variable μ_i . Substituting (29) and the constraint $m_i = c_i$ into the stage cost function $l(s_i, u_i)$ and the novel impedance model (13), we can obtain the transformed stage cost function $\tilde{l}_{c_i}(s_i, \mu_i)$ and the transformed novel impedance model $s_{i+1} = g_{c_i}(s_i, \mu_i)$. It should be noted that c_i is determined and known for all $i \in [t, t+H-1]$, and thus it does not serve as a variable. Then, denoting $\Psi_t = (\mu_t, \dots, \mu_{t+H-1})$, we can simplify the original optimization problem (26) and (27) as

$$\min_{\Psi_t} J(\Psi_t) = \sum_{i=t}^{t+H-1} l_{c_i}(s_i, \mu_i) + l_f(s_{t+H}) \quad (30)$$

subject to

$$\begin{cases} s_{i+1} = g_{c_i}(s_i, \mu_i) \\ s_i \in S_i \end{cases} \quad (31)$$

for all $i \in [t, t+H-1]$.

We then introduce the barrier function technique to soften the state constraints $s_i \in S_i$. The barrier function $\eta(s_i)$ is defined as

$$\eta(s_i) = \alpha(1 - \exp(-\phi(s_i, S_i)/\delta)) \quad (32)$$

where α and δ are positive scalars; function $\phi(s_i, S_i)$ represents the distance between the state s_i and the constrained set S_i . Incorporating the barrier function (32) into the novel stage cost function $\tilde{l}_{c_i}(s_i, \mu_i)$ and terminal cost function $\tilde{l}_f(s_{t+H})$, we can obtain the following softened optimization problem:

$$\min_{\Psi_t} J(\Psi_t) = \sum_{i=t}^{t+H-1} \tilde{l}_{c_i}(s_i, \mu_i) + \tilde{l}_f(s_{t+H}) \quad (33)$$

subject to

$$s_{i+1} = g_{c_i}(s_i, \mu_i) \quad (34)$$

for all $i \in [t, t+H-1]$ where

$$\begin{cases} \tilde{l}_{c_i}(s_i, \mu_i) = l_{c_i}(s_i, \mu_i) + \eta(s_i) \\ \tilde{l}_f(s_{t+H}) = l_f(s_{t+H}) + \eta(s_{t+H}). \end{cases} \quad (35)$$

Then, the simplified optimization problem (33) and (34) can be solved using the iterative linear quadratic regulator (iLQR) algorithm [29]. Once we compute the μ_t^* , we can compute the optimal stiffness and damping using the definition (29).

V. TANK-BASED APPROACH TO ENSURE SYSTEM PASSIVITY

This section is divided into two parts where in the first part we provide a passivity analysis for the continuous variable impedance system. However, every practical implementation is conducted in the discrete manner. Thus, in the second part we introduce how the passivity conditions derived in the first part can be used in the discrete control scenarios, such as MPC-based VIC approaches presented in the last section.

A. Passivity Analysis

We here utilize the tank-based approach to ensure the passivity of variable impedance system. The central idea of the tank-based approach is to store the energy dissipated by the variable impedance system in a tank and extract the stored energy from the tank for implementing the nonpassive variable impedance behaviors. We consider the following impedance system with variable damping and stiffness:

$$m\ddot{e} + d(t)\dot{e} + k_c e + k_v(t)e = f \quad (36)$$

where k_c and $k_v(t)$ are the constant and variable parts of the stiffness, respectively. The damping $d(t)$ and stiffness $k_c + k_v(t)$ are always positive for any t . The constant part of stiffness can be arbitrarily designed; for example, we can set it as zero if it is not needed.

Let us firstly consider the following storage function of the variable impedance system (36):

$$V = \frac{m}{2}\dot{e}^2 + \frac{k_c}{2}e^2. \quad (37)$$

Its time-derivative can be computed as

$$\begin{aligned} \dot{V} &= m\dot{e}\ddot{e} + k_c e\dot{e} \\ &= m\dot{e} \frac{f - k_v(t)e - k_c e - d(t)\dot{e}}{m} + k_c e\dot{e} \\ &= \dot{e}f - (d(t)\dot{e}^2 + k_v(t)e\dot{e}). \end{aligned} \quad (38)$$

The above equation indicates that the rate of dissipated energy by the variable impedance system is $d(t)\dot{e}^2 + k_v(t)e\dot{e}$. Note that the term “dissipated” here has its direction, i.e., if $d(t)\dot{e}^2 + k_v(t)e\dot{e} < 0$, the energy is negatively dissipated and actually injected to the variable impedance system (36). A tank is used to accept the dissipated energy. However, the volume of the tank is limited and it also should not be exhausted. Thus, the energy variation of the tank is designed as

$$\dot{\Gamma}(t) = \sigma_1 (d(t)\dot{e}^2 + k_v(t)e\dot{e}) \quad (39)$$

where $\Gamma(t)$ is the tank energy; parameter σ_1 is used to ensure that the tank energy in a predefined interval $[\Gamma_{\min}, \Gamma_{\max}]$, $\Gamma_{\min} > 0$ and it is designed as

$$\begin{cases} \sigma_1 = 0, d(t)\dot{e}^2 + k_v(t)e\dot{e} > 0 & \text{and } \Gamma > \Gamma_{\max} \\ \sigma_1 = 0, d(t)\dot{e}^2 + k_v(t)e\dot{e} < 0 & \text{and } \Gamma < \Gamma_{\min} \\ \sigma_1 = 1. \end{cases} \quad (40)$$

The above definition implies that the energy channel between the tank and the variable impedance system will be closed if the condition $\Gamma \in [\Gamma_{\min}, \Gamma_{\max}]$ is going to be violated. We have introduced how to store the energy into a tank; now, we introduce how to use the stored energy to implement variable impedance behaviors. From (38), we can find that only the variable stiffness may inject energy into the variable impedance system, thus we introduce a condition to the stiffness variation, which raises the following conditioned variable impedance system:

$$m\ddot{e} + d(t)\dot{e} + k_c e + \sigma_2 k_v(t)e = f \quad (41)$$

where the condition variable σ_2 is designed as

$$\begin{cases} \sigma_2 = 1, d(t)\dot{e}^2 + k_v(t)e\dot{e} > 0 \\ \sigma_2 = \sigma_1, \text{else.} \end{cases} \quad (42)$$

The above condition implies that if the energy is positively dissipated, i.e., $d(t)\dot{e}^2 + k_v(t)e\dot{e} > 0$, the variable stiffness behavior will be fully conducted. Otherwise, the variable stiffness behavior will depend on the tank parameter σ_1 . We will show that the conditioned variable impedance system (41) is passive with respect to the input–output pair $(f, \dot{e})$.

Theorem 1: The conditioned variable impedance system (41) is passive with respect to the input–output pair $(f, \dot{e})$ if the tank dynamic $\dot{\Gamma}$, tank parameter σ_1 , and the condition variable σ_2 are designed as (39), (40), and (42), respectively.

Proof: Considering the following storage function:

$$W = \frac{m}{2}\dot{e}^2 + \frac{k_c}{2}e^2 + \Gamma \quad (43)$$

its time-derivative can be computed as

$$\begin{aligned} \dot{W} &= m\dot{e}\ddot{e} + k_c e\dot{e} + \dot{\Gamma} \\ &= \dot{e}f - \sigma_2 k_v(t)e\dot{e} - d(t)\dot{e}^2 + \sigma_1 (d(t)\dot{e}^2 + k_v(t)e\dot{e}) \\ &= \dot{e}f - (1 - \sigma_2)d(t)\dot{e}^2 - (\sigma_2 - \sigma_1) (d(t)\dot{e}^2 + k_v(t)e\dot{e}) \end{aligned} \quad (44)$$

where the second line is obtained by substituting the conditioned variable impedance system (41) and tank dynamic (39) into the first line. According to the definitions of σ_1 and σ_2 in (40) and (42), respectively, we can conclude that

$$\begin{cases} \sigma_2 \leq 1 \\ (\sigma_2 - \sigma_1) (d(t)\dot{e}^2 + k_v(t)e\dot{e}) \geq 0. \end{cases} \quad (45)$$

Substituting (45) into the last line of the (44) and using the fact that $d(t) > 0$, we can conclude that

$$\dot{W} \leq \dot{e}f \quad (46)$$

which implies the passivity condition

$$W(t) - W(0) \leq \int_0^t \dot{e}(\tau)f(\tau)d\tau \quad (47)$$

and completes the proof. $\square$

B. Practical Implementation

In the last section, we have introduced an approach to ensure the passivity of the continuous variable impedance system. However, to be practical, we must implement the approach in the discrete manner. Before introducing main contents in this section, we make a basic assumption that conditions to ensure the passivity of the continuous system are still useful to ensure passivity of the corresponding discrete system if the control period T used in discretization process is small enough. Actually, this assumption may be the case for many control engineering practices where the system stability is analyzed in the continuous manner.

It should be noted that the key of the implementation is to determine the condition parameter σ_2 . For practical implementation, we care about how to compute the σ_2 at current time step based on the current information. Available information at time step i contains $(\Gamma_i, e_i, \dot{e}_i)$, and we can firstly compute d_i and $k_{v,i}$ using the presented MPC approaches. According to (40), we can then compute the time step i 's tank parameter $\sigma_{1,i}$, and further compute the condition parameter $\sigma_{2,i}$ according to (42). Arriving here we can implement the control according to (41) as

$$v_i = \ddot{\xi}_i = \frac{f_i - (k_c + \sigma_{2,i}k_{v,i})e_i - d_i\dot{e}_i}{m} + \ddot{\xi}_{r,i} \quad (48)$$

where v_i will be sent to the low level controller (4). Finally, the next-time tank energy Γ_{i+1} can be computed by integrating $\dot{\Gamma}_i$ as $\Gamma_{i+1} = \Gamma_i + \dot{\Gamma}_i T$. After a control period T , the information $(e_{i+1}, \dot{e}_{i+1})$ will be automatically generated by the real environment, which completes an iteration. The specific algorithm for the proposed MPC-based VIC approach is concluded as Algorithm 1.

VI. EXPERIMENTS

This section is divided into two parts, where contact-rich and contact-free experiments are conducted, respectively, to validate the proposed MPC-based VIC algorithm. Two contact-rich experiments are designed where the first one is used to validate the OS MPC-based VIC's capacity for online adjustment of the precision-compliance tradeoff modes. The second contact-rich experiment focuses on validating the state constraints capacity of the MS MPC-based VIC.

In the contact-free experiment, we show that the proposed MPC-based VIC algorithm will behave properly under different reference trajectories, i.e., have large and low impedance parameters for high-velocity and low-velocity reference trajectories, respectively. We will also provide an approach to make the algorithm pay more attention to increase tracking precision. All the experiments are conducted on the Franka-Panda robot

Algorithm 1: The Proposed MPC-Based VIC Approach.

for Time step $i = 1, \dots$ **do**
 Observing the state s_i
 Computing the original variable impedance law
 ($d_i, k_{v,i}$) by solving the MPC optimization problem
 (23), (24) or (33), (34) according to whether the state
 constraints are also considered
 Computing the tank parameter $\sigma_{1,i}$ and condition
 parameter $\sigma_{2,i}$ according to (40) and (42), respectively.
 Computing the desired acceleration according to (48)
 Computing the torque according to the low level
 controller (4) and sending it to the manipulator
 Computing the tank energy variation $\dot{\Gamma}_i$ according to
 (39)
 Computing the next-time tank energy as
 $\Gamma_{i+1} = \Gamma_i + \dot{\Gamma}_i T$
end for



Fig. 2. First contact-rich experiment.

and a computer with Intel Core i7-8700 K CPU, 3.70 GHz. The VIC approaches are achieved using the python language with packages numpy and cvxopt. The low level control (4) is achieved using the C++ libfranka.

A. Contact-Rich Experiments

1) *First Contact-Rich Experiment:* The first contact-rich experiment is shown in Fig. 2 where the robot is asked to track the reference trajectory (red colored in Fig. 2) under contact with curve obstacles. In experiments of this article, the contact force is measured using a 6-D force/torque sensor. Curve obstacles provides specific shape spaces (as the wavy curve shape space shown in Fig. 2) in which the robot can move freely, otherwise the robot motion will be prevented and reshaped. A fully compliant robot will be easily influenced by the contact environment. For example, it may stop at a specific point because of the friction and resistance provided by the obstacles. By contrast, a precision robot (such as a robot under position control) will try its best to track the reference trajectory even its behaviors will generate large interactive forces. Our purpose is to adaptively adjust the precision-compliance tradeoff mode to make the robot track the reference trajectory well and maintain a certain degree of compliance. The OS MPC-based VIC approach introduced in Section III-A is used to achieve this purpose. We choose over-damping impedance constraints to conduct analyses. Control and constraint parameter settings for the OS MPC-based VIC approach are listed as $k_{\min,t} = 100$ N/m, $k_{\max,t} = 1500$ N/m, $c_t = 1$, $Q = \text{diag}(50, 0.1, 0.1)$, $R = \text{diag}(10^{-6}, 10^{-6}, 10^{-6})/500$, $T = 0.015$ s, $k_c = 300$ N/m, $\Gamma_0 = 0.1$, $\Gamma_{\max} = 1$, and $\Gamma_{\min} = 0.01$

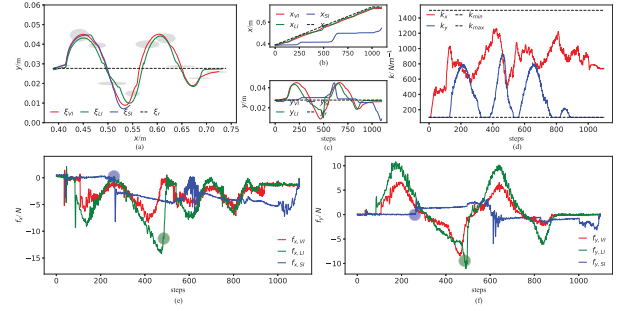


Fig. 3. Comparison results for the OS MPC-based VIC, small impedance control, and large impedance control. (a) Reference and real trajectories of different controllers. (b) Reference and real x-trajectories of different controllers. (c) Reference and real y-trajectories of different controllers. (d) Stiffness trajectories for the OS MPC-based VIC approach. (e) f_x trajectories of different controllers. (f) f_y trajectories of different controllers.

where $\text{diag}(\cdot)$ represents a diagonal matrix with components listed in the brackets. Sampling period is equal to the control period T in this article. It should be noted that VIC is only conducted on the (x, y) plane, and the z -direction is applied force control, which is presented in [28]. We compare the experimental results of the proposed OS MPC-based VIC approach with these of the large impedance control ($k_t = 1500$), small impedance control ($k_t = 100$), classical MPC-based VIC [20], [21], and RL-based VIC [16] approaches. Large and small impedance control approaches are served as baselines, and they can be regarded as precision position control and fully compliant control, respectively.

To make the results clearer, we only plot the comparative results between the proposed MPC-based VIC, large impedance control, and small impedance control approaches, and leave the full comparative results to be shown by bar charts with accuracy and compliance performance measures. Comparison results between the proposed MPC-based VIC, large impedance control, and small impedance control approaches are shown in Fig. 3. Fig. 3(a)–(c) shows the reference trajectories and real trajectories generated by the OS MPC-based VIC, large impedance control, and small impedance control. Subscripts VI , LI , SI , and r denote the OS MPC-based VIC, large impedance control, small impedance control, and reference respectively. As we can see, the VIC and large impedance control can both allow the robot to complete the tracking task. However, the robot motion will be prevented if it is controlled by the small impedance controller. These phenomena are reasonable because the large and small impedance control pay attention to precision and compliance, respectively, and the proposed VIC approach focuses on searching proper tradeoffs between precision and compliance. The central idea of the proposed VIC is to generate the large/small stiffness parameter for the large/small tracking error. This idea is directly shown by the stiffness ellipses in Fig. 3(a). The height and width of the stiffness ellipse represent the stiffness values in the y and x directions, respectively. We can find that k_y increases as $|e_y|$ increases. The specific stiffness trajectories in the x and y directions are shown in Fig. 3(d). As we can see, the value of k_x is always larger than the value of k_y , which is caused by the

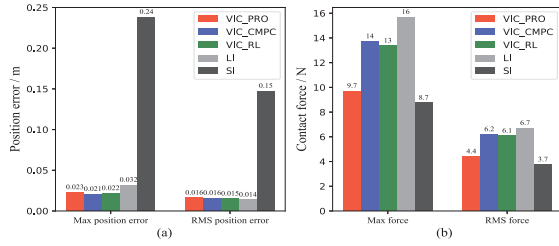


Fig. 4. Quantitative descriptions of the accuracy and compliance performance for the proposed MPC-based VIC, classical MPC-based VIC, RL-based VIC, large impedance control, and small impedance control approaches. (a) Bar chart for accuracy performance. (b) Bar chart for compliance performance.

fact that $|e_y|$ is smaller than $|e_x|$. This fact can be obtained by comparing the scales of the abscissa and ordinate of Fig. 3(a). Moreover, stiffness parameters always satisfy the impedance constraints, which implies that we can moderate values of $k_{\min}$ and $k_{\max}$ to artificially determine the extreme robot precision and compliance properties. Fig. 3(e) and (f) shows the contact forces in the x and y directions, respectively. As is expected, the contact force of the large/small impedance control is the highest/lowest, and the force of VIC is in between. A very interesting thing shown in Fig. 3(e) and (f) is that the large impedance control and small impedance control both raise jump behaviors, such as at approximately step 485 (indicated by the green circle)/step 260 (indicated by the blue circle) for large/small impedance control. These jump behaviors can also be observed at Fig. 3(a)–(c). Jump behaviors of small and large impedance control are caused by the cumulative tracking error and large stiffness, respectively. The proposed VIC can handle these issues by online adjusting the precision-compliance tradeoff modes, and thus, the motion and force trajectory shapes of the VIC are smoother.

Remark 2: The static errors in Fig. 3(a)–(c) is caused by the static friction. Large stiffness can decrease the influence of the friction, and thus, have the smaller static error.

The bar charts showing full comparative results are given in Fig. 4 where labels “VIC_PRO,” “VIC_CMPC,” “VIC_RL,” “LI,” and “SI” represent the proposed MPC-based VIC, classical MPC-based VIC, RL-based VIC, large impedance control, and small impedance control approaches, respectively. We choose the maximum/root-mean-square position error and maximum/root-mean-square force to measure the accuracy and compliance performances, respectively. As is expected, the large and small impedance control act as two extremes, and these VIC approaches can both improve the overall performance. As we can see, these VIC approaches can decrease the position error into a similar level with that of the large impedance control approach, while maintain a better compliance performance compared with the large impedance control approach. However, both classical MPC-based VIC and RL-based VIC approach have obviously worse compliance performance than that of the small impedance control approach. This is because that the classical MPC cannot directly handle the accuracy-compliance tradeoff problem as mentioned in Section III, and the RL algorithm always suffers from the model gap and local optimum problems. On the contrary, the proposed MPC-based VIC approach

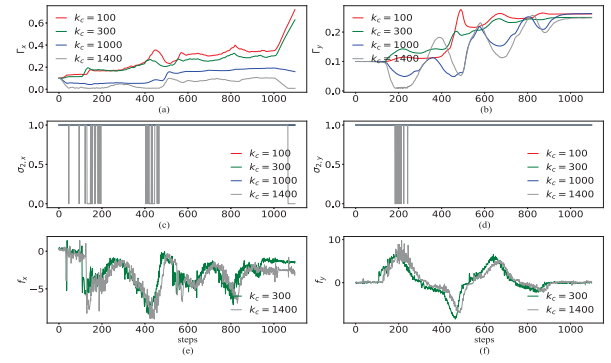


Fig. 5. Tank energy and parameter variations of the proposed MPC-based VIC approach with different settings of k_c . (a) Tank energy variation in x direction with different k_c . (b) Tank energy variation in y direction with different k_c . (c) σ_2 trajectory in x direction with different k_c . (d) σ_2 trajectory in y direction with different k_c . (e) f_x trajectories under different k_c settings. (f) f_y trajectories under different k_c settings.

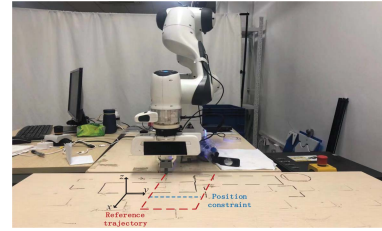


Fig. 6. Second contact-rich experiment.

performs similarly with the small impedance control approach with respect to the compliance measure, which implies that the proposed MPC-based VIC approach can properly and adaptively handle the accuracy-compliance tradeoff problem and ensure the accuracy and compliance simultaneously.

We also investigate the influence of the tank-based approach. Tank energy and parameter variations with different k_c settings are shown in Fig. 5. As we can find in Fig. 5(a)–(d), the range of k_c to maintain the original variable impedance law derived by the MPC approach is very large, which is reflected by the fact that only k_c is set as a very large value 1400 (approach the maximum value 1500), the tank could have the probability to be exhausted ($\Gamma = \Gamma_{\min} = 0.01$), and might make $\sigma_2 = 0$ according to (40) and (42). We also investigate the influence of the suddenly switch of σ_2 . It can be seen from Fig. 5(e) and (f) that there exist high-frequency force variations around the σ_2 -switch areas (step 50–step 200, step 400–step 450 for f_x , and step 190–step 210 for f_y). These force tremors are caused by the high-frequency stiffness variations due to the high-frequency σ_2 switches. Although these tremors will not have very serious influences with respect to the performance measures according to the experiment results, we recommend to avoid these tremors by choosing the k_c in a proper range.

2) Second Contact-Rich Experiment: The second contact-rich experiment is shown in Fig. 6 where the robot is asked to track a reference trajectory (red colored) with the position constraint (blue colored) under the control of the MS MPC-based VIC. Comparative experiments are not conducted here since

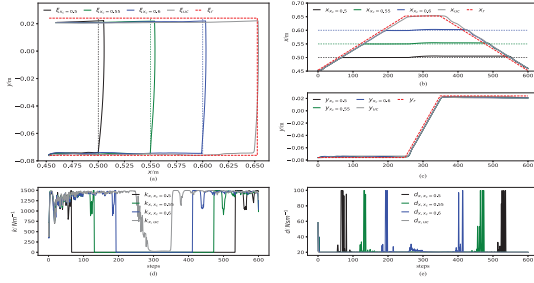


Fig. 7. Experimental results of VIC with different position constraints. (a) Reference and real trajectories of VIC with different position constraints. (b) Reference and real x -trajectories of VIC with different position constraints. (c) Reference and real y -trajectories of VIC with different position constraints. (d) Stiffness trajectories of VIC with different position constraints. (e) Damping trajectories of VIC with different position constraints.

both classical MPC-based VIC and RL-based VIC approaches cannot handle state constraints. For the position constraint shown in Fig. 6, the distance function $\phi(s_i, S_i)$ in (32) can be designed as

$$\phi(s, S) = \begin{cases} 0, [1 & 0 & 0] s + x_r \leq x_c \\ [1 & 0 & 0] s + x_r - x_c, \text{else} \end{cases} \quad (49)$$

where x_c represents the constrained position in the x direction. Other control and constrained parameters are set as $k_{\min,t} = 0$, $k_{\max,t} = 1500$, $d_{\min,t} = 20$, $d_{\max,t} = 100$, $c_t = 1$, $Q = \text{diag}(50, 0.1, 0.1)$, $R = \text{diag}(10^{-6}, 10^{-6}, 10^{-6})/1000$, $T = 0.015$, $H = 10$, $\delta = 2$, and $\alpha = 50$.

Remark 3: We choose the position constraint as a specific state constraint for intuitive experimental effect. Practically, other state constraints such as velocity constraint can also be achieved using the MS MPC-based VIC approach.

Experimental results of MS MPC-based VIC with different position constraints are shown in Fig. 7, where subscripts indicate the values of different position constraints ('uc' indicates that the algorithm is conducted without any position constraint and can be used for a benchmark). From Fig. 7(a) and (b), we can find that all position constraints (represented by dot lines with different colors) can be approximately satisfied with slight violations. These slight violations may be caused by various reasons. Firstly, the low-level controller may be imprecision due to the inaccurate model parameters. Secondly, the constrained optimization (26) and (27) is solved using the barrier function technique, which belongs to soften techniques those can reduce computation burden but with the expense of losing the hard constraint property. Thus, slight constraint violations are acceptable for the optimization algorithm. Fig. 7(d) and (e) shows the ways how the MS MPC-based VIC to ensure these position constraints. As we can see, when the robot approaches the constrained areas, the stiffness and damping parameters will be suddenly changed to $k_{t,\min}$ and $d_{t,\max}$, respectively. These phenomena are reasonable since the small stiffness will reduce the spring-like force, which will make the robot track the reference trajectory lying in the constraint-violated area [as shown in Fig. 7(a) and (b)]. On the contrary, the large damping parameters will increase the viscosity of the system, which will prevent robot motions. The precondition for this property is

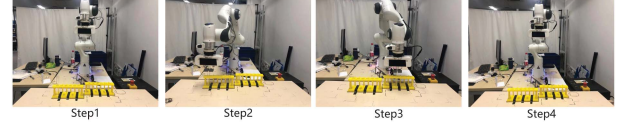


Fig. 8. Contact-free experiment.

that the reference velocity is set as zero, which will lead to a classical mass-damping-spring system. In this experiment, $k_{t,\min}$ and reference velocity are all set as zero, and thus, when the robot approaches the constraint-violated area, we can totally remove the spring-like force (the power to drive the robot to enter the constraint-violated area) and increase the system viscosity (the power to prevent robot motions).

B. Contact-Free Experiment

The previous contact-rich experiments show the proposed VIC approach's capacities for online precision-compliance tradeoff adjustment and handling state-dependent constraints. In this section, we focus on contact-free tasks and discuss how the precision-compliance tradeoff adjustment capacity of the proposed VIC approach can benefit the contact-free cases. The experimental design is shown in Fig. 8. During the step1 to step2, the robot is asked to track the first reference trajectory to arrive a posture where the robot can clamp a test-tube. During the step2 to step3, the robot is asked to track the second reference trajectory to insert the test-tube into a hole of the test-tube support. During the step3 to step4, the robot is asked to track the third reference trajectory to arrive a predefined posture. We refer to this experiment as "contact-free" experiment since no contact is expected to be happened during the experiment. The OS MPC-based VIC approach is used for these trajectory tracking tasks.

Different from previous contact-rich experiments, this contact-free experiment more highly focuses on the tracking accuracy, especially for the second trajectory tracking processes. If the static error is too large, the test-tube will not be inserted into the hole and there may raise a large contact force caused by the collision. There are many approaches to increase the tracking accuracy, such as increasing Q and decreasing R . However, a more convenient and intuitive way is to change the $k_{t,\min}$ according to the current robot state. As discussed in Section IV-A, a state-dependent $k_{t,\min}$ will not influence the QP form of the OS MPC optimization problem. To increase tracking accuracy, the state-dependent $k_{t,\min}$ is designed as

$$k_{t,\min} = (k_{t,\max} - \bar{k}_{\min} - \bar{\delta}) \exp\left(-\frac{\|\xi_t - \xi_{\text{goal}}\|_2^2}{2\sigma^2}\right) + \bar{k}_{\min} \quad (50)$$

where $\bar{k}_{\min}$ and $\bar{\delta}$ are two constants, and they represent the minimal value of $k_{t,\min}$ and the distance between the largest value of $k_{t,\min}$ and $k_{t,\max}$, respectively. The above function determines a bell curve and implies that $k_{t,\min}$ will achieve its largest value $k_{t,\max} - \bar{\delta}$ when the current posture ξ_t equals to the goal posture ξ_{goal} . Besides, the sharpness of the bell curve is determined by the parameter σ . In this experiment, we set $\bar{k}_{\min} = 100$, $\bar{\delta} = 50$, and $\sigma = 0.08$. Other parameters for OS MPC-based

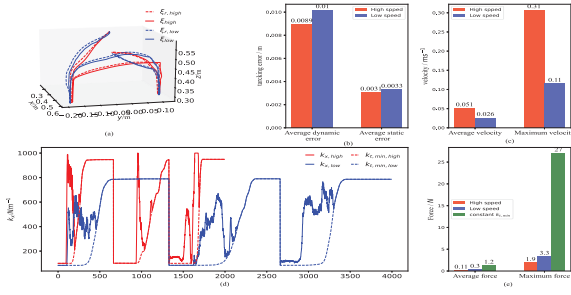


Fig. 9. Experimental results of VIC with high and low speed reference trajectories. (a) Reference and real trajectories of VIC with high and low speed reference trajectories. (b) Bar chart for tracking error. (c) Bar chart for velocity. (d) k_x trajectories of VIC with high and low speed reference trajectories. (e) Bar chart for contact force.

VIC is set as $k_{\max,t} = 1000$, $c_t = 1$, $Q = \text{diag}(50, 0.1, 0.1)$, $R = \text{diag}(10^{-6}, 10^{-6}, 10^{-6})/1000$, and $T = 0.015$ s.

It is expected that using the OS MPC-based VIC approach, the robot can adaptively tune its impedance parameters to properly track different types of reference trajectories for contact-free tasks. Concretely, the robot is expected to have the large and low stiffness to track the high and low speed reference trajectories, respectively. This variable mode is desirable since the robot is always expected to complete the tracking task with impedance parameters, or gains as small as possible. This setting also reflects the variable precision-compliance tradeoff modes although there may be no contact in the contact-free tasks.

To validate this capacity of the OS MPC-based VIC approach, we demonstrate the robot with two types of reference trajectories, i.e., high speed and low speed reference trajectories, to complete the task shown in Fig. 8. Comparison results are shown in Fig. 9 where subscripts “low” and “high” are used to distinguish the low and high speed cases, respectively. From Fig. 9(a) and (b), we can find that both tracking tasks can be well completed with the small dynamic error and tiny static error. The dynamic error is small since we choose a relatively small R , and the static error is approximately eliminated using the law (50), which can be seen in Fig. 9(d). As we can see, values of $k_{t,min,low}$ and $k_{t,min,high}$ both gradually increase when the robot approaches the goal position, which will generate stiff robotic behaviors near the robot goal position and ensure a precision control (similar with position control). When the distance between the robot position and goal position is large, the law (50) can be neglected and the robot is controlled by the VIC, which also considers the robot compliance. We then compare the behaviors between the low and high speed cases in the area away from the goal position. As we can see, the shape of stiffness trajectories in the high-speed cases is sharper than that in the low-speed cases. This phenomenon is not only caused by the fact that the tracking time for the high-speed case is shorter but also due to the larger stiffness chosen in the high-speed case. Fig. 9(e) shows the contact force in the z direction, which can indicate whether the test-tube is well inserted into the hole. The additional label “constant $k_{t,min}$ ” represents the f_z corresponding to the high-speed case where the law (50) is not used. As what we expected, the test-tube can

TABLE I
PROPERTY COMPARISON OF THE PROPOSED MPC-BASED VIC APPROACH AND OTHER EXISTED APPROACHES

	Prop_VIC	CMPC_VIC [20], [21]	RL_VIC [16]
Passivity	✓	✗	✗
Imp. con. (TV)	✓(✓)	✗(✗)	✓(✗)
State con. (TV)	✓(✓)	✗(✗)	✗(✗)

be successfully and smoothly inserted into the hole if we apply the state-dependent law (50) for $k_{t,min}$, which can be reflected by the small contact forces. On the contrary, when the law (50) is not used, the test-tube will not be aligned with the hole, and thus, a large collision force will be raised.

VII. DISCUSSION

In this section, we give a more intuitive discussion for advantages of the proposed MPC-based VIC approaches over other existent approaches. Properties of these approaches are listed in Table I where “Prop_VIC,” “CMPC_VIC,” and “RL_VIC” stand for the proposed MPC-based VIC, classical MPC-based VIC, and RL-based VIC approaches, respectively. “TV” is the abbreviation of time-varying. As we can see, only the proposed VIC approach can ensure the system passivity. For constraints, the classical MPC-based VIC approach cannot handle any constraints, whether the constraint is time-varying or time invariant. The RL-based VIC can only handle time-invariant impedance constraints utilizing the special structure of the neural network (such as tanh activation function). The proposed MPC-based approach can handle any constraints.

VIII. CONCLUSION AND FUTURE WORKS

This article presented novel MPC-based VIC approaches to deal with the adaptive precision-compliance tradeoff problem for robot tracking tasks. A novel impedance model was established, which allows us to consider impedance and state constraints during the MPC-based VIC design. Impedance constraints are significantly useful to ensure some extreme robot properties such as worst tracking precision (by tuning $k_{t,min}$) and worst contact compliance (by tuning $k_{t,max}$). Two types of MPC approaches, i.e., OS MPC and MS MPC approaches, were presented to design variable impedance laws under only impedance constraints and both impedance and state constraints, respectively. We also provided a tank-based approach to ensure the passivity of the variable impedance system. We validated the proposed VIC approaches by two types of tasks, i.e., contact-rich and contact-free tasks.

In this article, the dynamic model uncertainty of the robot is not considered, which might reduce the control performance. Thus, VIC approaches with online adaptive model learning will be researched in our future works.

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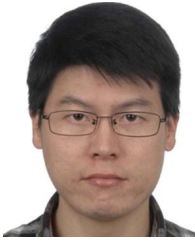
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